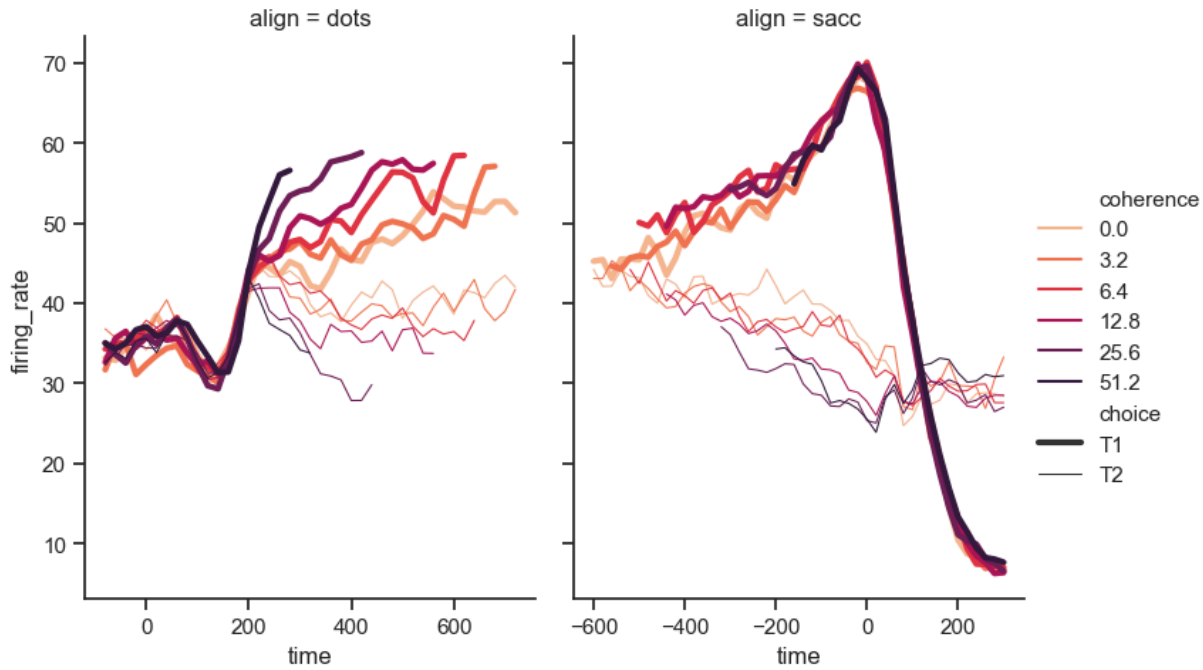
In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

kashti = sns.load_dataset("dots")
kashti.head()
```

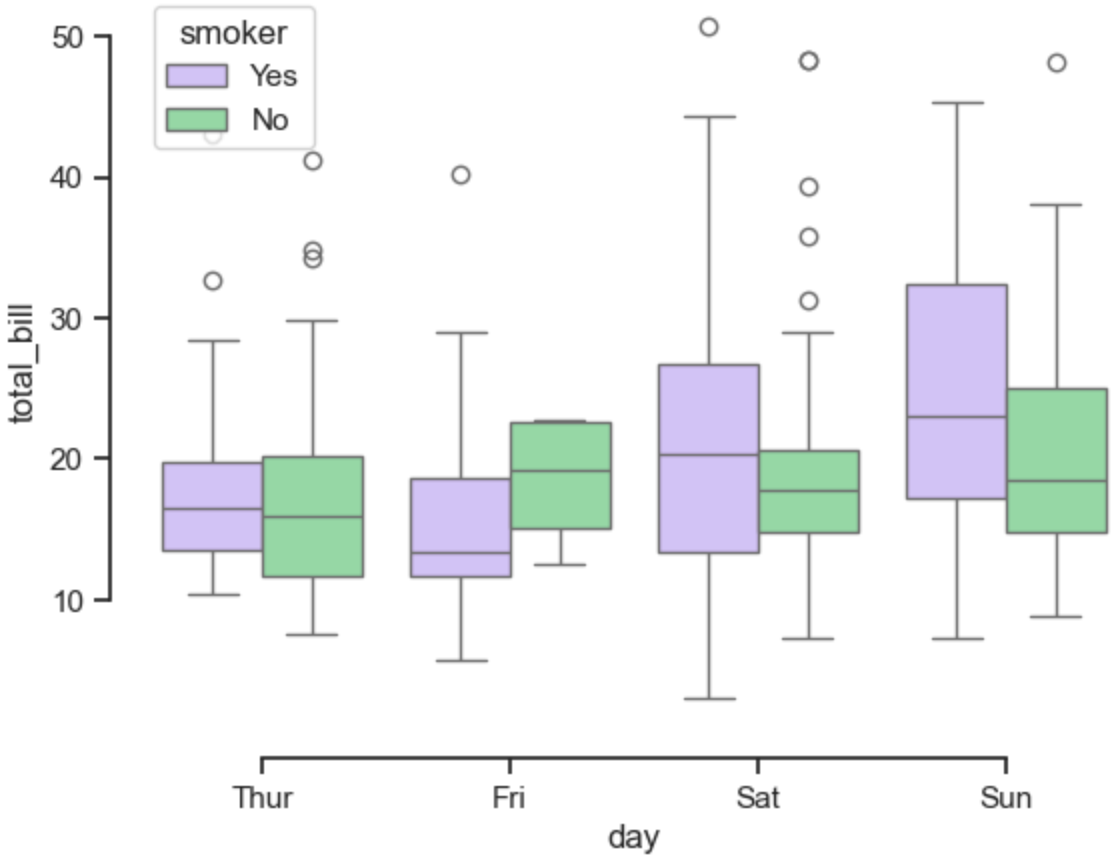
Out[3]:

| | align | choice | time | coherence | firing_rate |
|---|---|---|---|---|---|
| **0** | dots | T1 | -80 | 0.0 | 33.189967 |
| **1** | dots | T1 | -80 | 3.2 | 31.691726 |
| **2** | dots | T1 | -80 | 6.4 | 34.279840 |
| **3** | dots | T1 | -80 | 12.8 | 32.631874 |
| **4** | dots | T1 | -80 | 25.6 | 35.060487 |

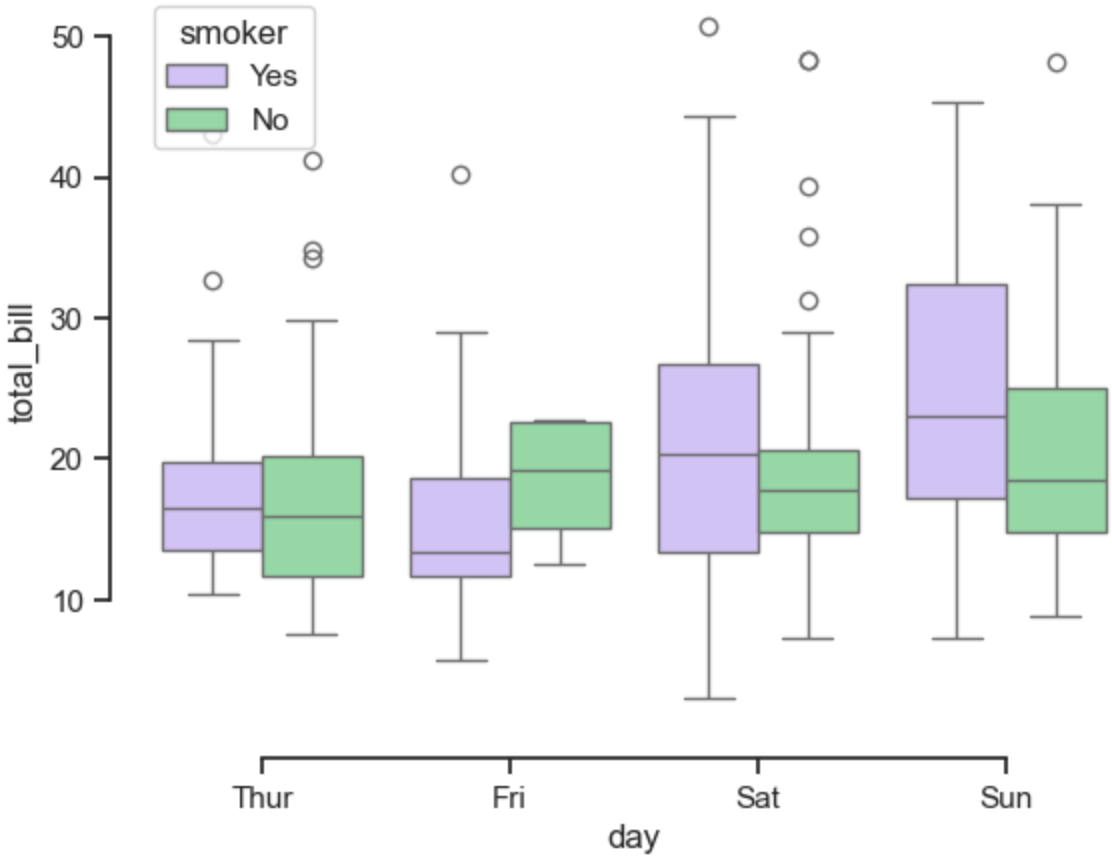
In [17]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

nuqta = sns.load_dataset("dots")

# defining a color palette
sns.set_theme(style="ticks")
#defining a color palette
p = sns.color_palette("rocket_r")

# plot line plot
sns.relplot(data=nuqta,
           x="time", y="firing_rate",
           hue="coherence",
           size="choice",
           col="align",
           kind="line",
           size_order=["T1","T2"],
           palette=p,
           height=5, aspect=.75, facet_kws=dict(sharex=False),)
plt.show()
```




In [25]:
```
import seaborn as sns
import matplotlib.pyplot as plt
sns.set_theme(style="ticks", palette="pastel")

#load the example tips dataste
tips = sns.load_dataset("tips")

# Draw a nested boxplot to show bills by day and time
sns.boxplot(x="day", y="total_bill",
            hue="smoker", palette=["m", "g"],
            data=tips)
sns.despine(offset=10, trim=True)
plt.show()
```

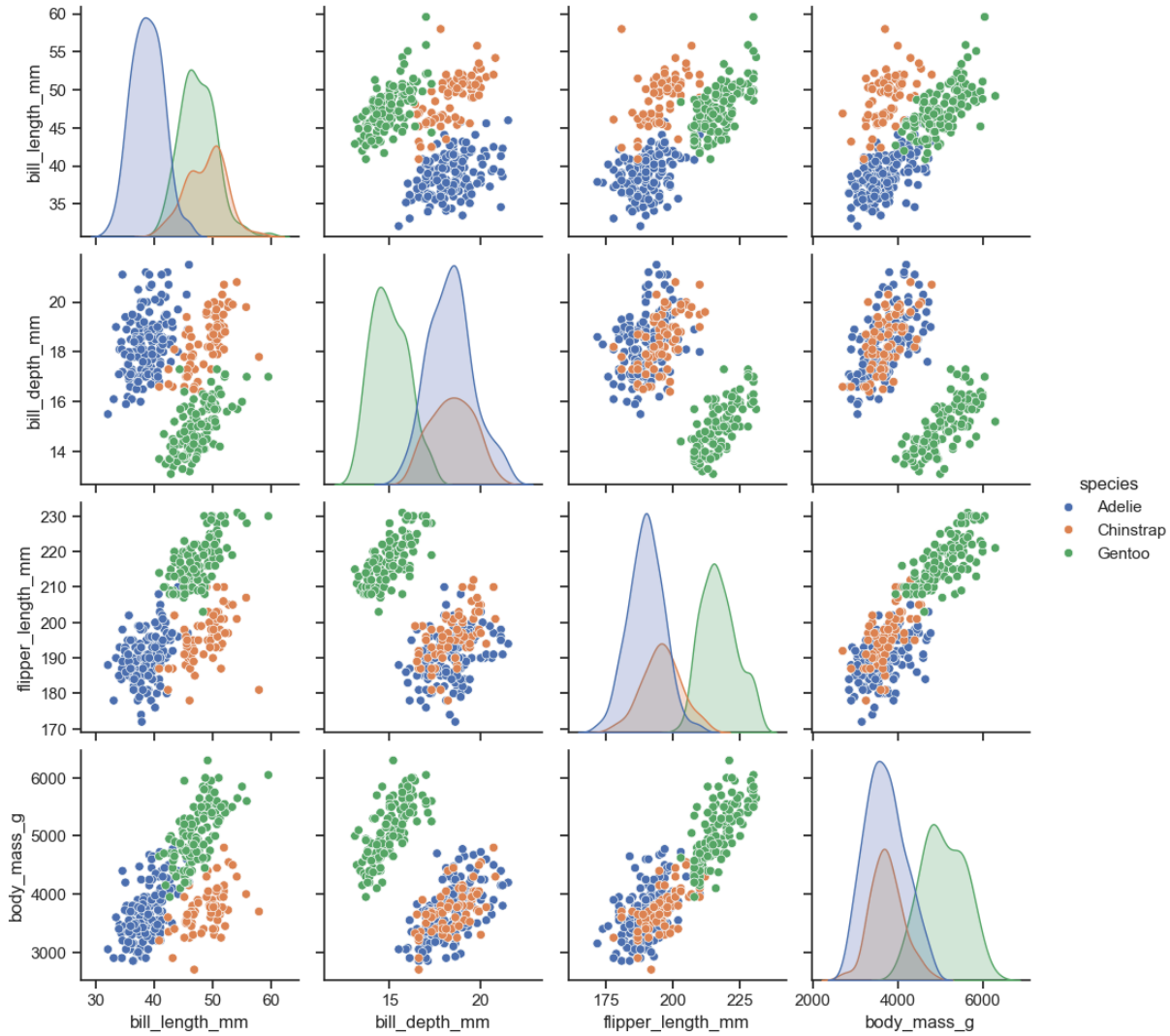



In [21]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style="ticks", palette="pastel")

# load the example tips dataset
tips = sns.load_dataset("tips")

# Draw a nested boxplot to show bills by day and time
sns.boxplot(x="day", y="total_bill",
            hue="smoker", palette=["m", "g"], # Fixed the spelling here
            data=tips)

sns.despine(offset=10, trim=True)
plt.show()
```




In [28]:
```
import seaborn as sns
sns.set_theme(style="ticks")
df = sns.load_dataset("penguins")
sns.pairplot(df, hue="species")
```

Out[28]: <seaborn.axisgrid.PairGrid at 0x1a061687750>



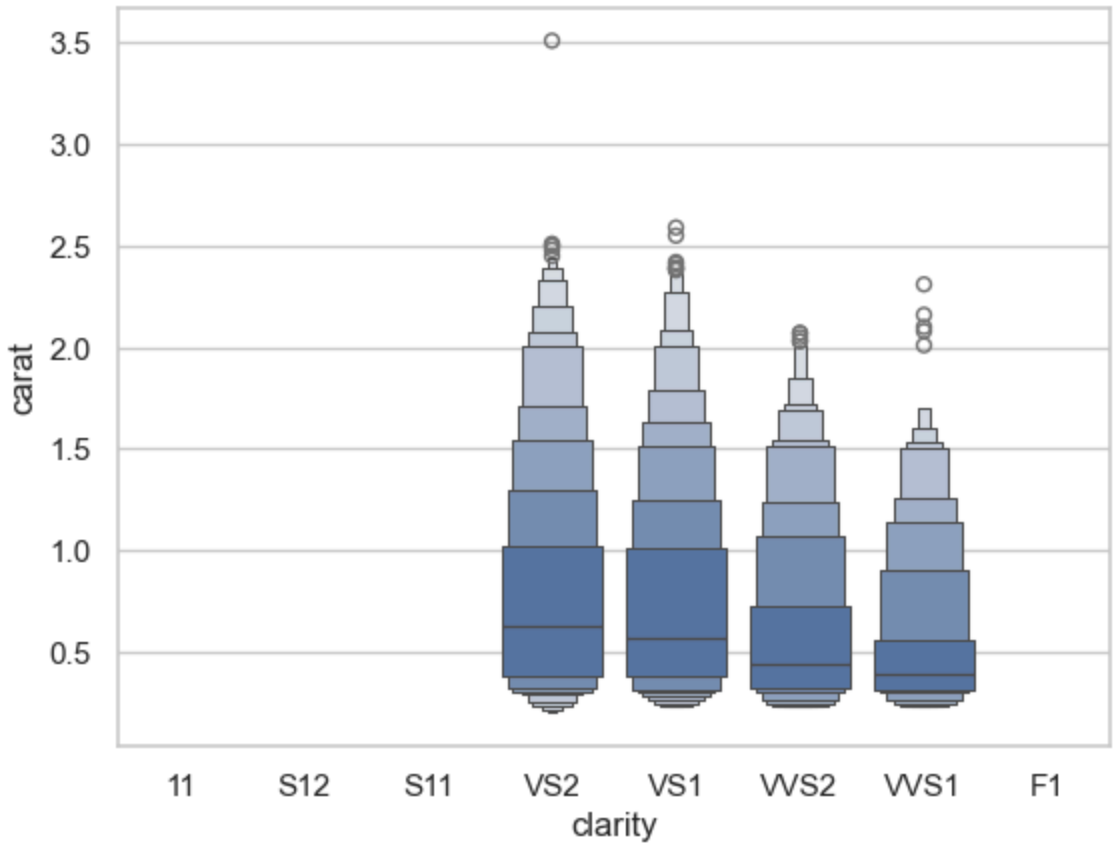

In [29]:
```
diamonds = sns.load_dataset("diamonds")
diamonds.head()
```

Out[29]:

| | carat | cut | color | clarity | depth | table | price | x | y | z |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 0.23 | Ideal | E | SI2 | 61.5 | 55.0 | 326 | 3.95 | 3.98 | 2.43 |
| **1** | 0.21 | Premium | E | SI1 | 59.8 | 61.0 | 326 | 3.89 | 3.84 | 2.31 |
| **2** | 0.23 | Good | E | VS1 | 56.9 | 65.0 | 327 | 4.05 | 4.07 | 2.31 |
| **3** | 0.29 | Premium | I | VS2 | 62.4 | 58.0 | 334 | 4.20 | 4.23 | 2.63 |
| **4** | 0.31 | Good | J | SI2 | 63.3 | 58.0 | 335 | 4.34 | 4.35 | 2.75 |

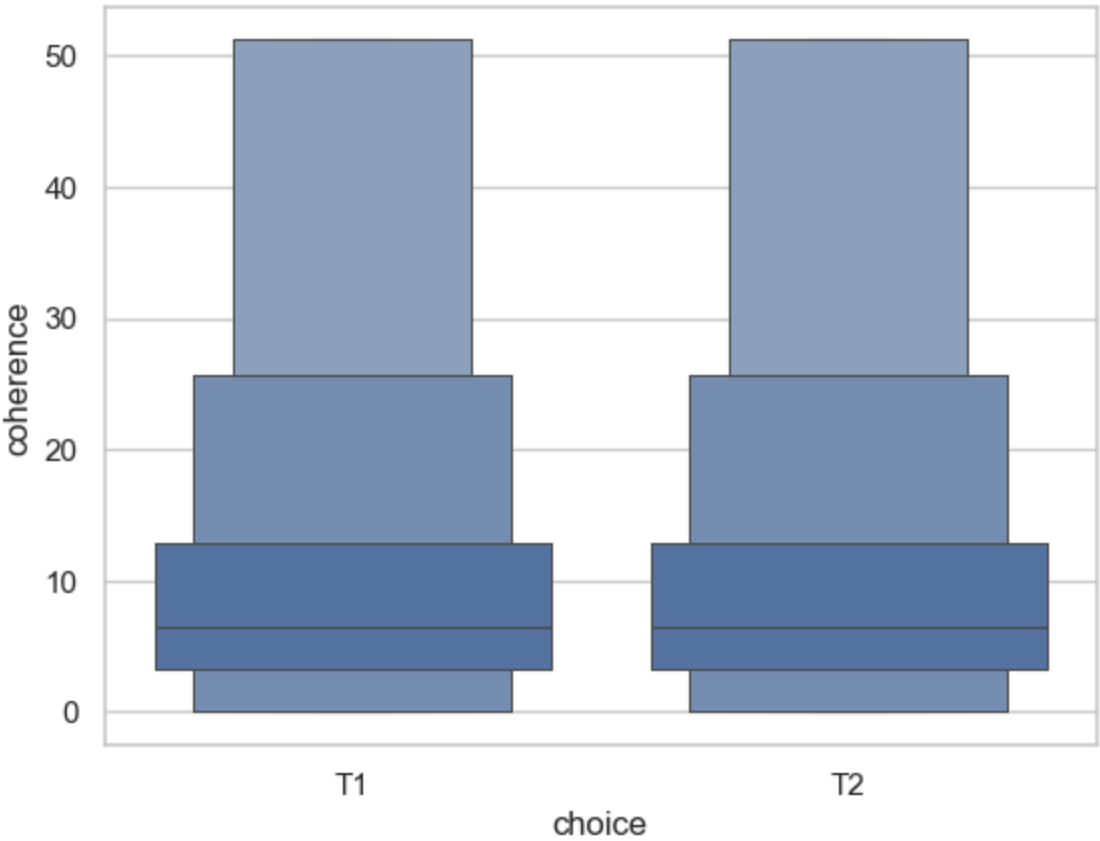
In [35]:
```
import seaborn as sns
sns.set_theme(style="whitegrid")

diamonds = sns.load_dataset("diamonds")
clarity_ranking = ["11", "S12","S11", "VS2","VS1", "VVS2", "VVS1", "F1"]
sns.boxenplot(x="clarity", y="carat",
              color="b", order=clarity_ranking, width_method="linear", data=diamonds
```

Out[35]: <Axes: xlabel='clarity', ylabel='carat'>



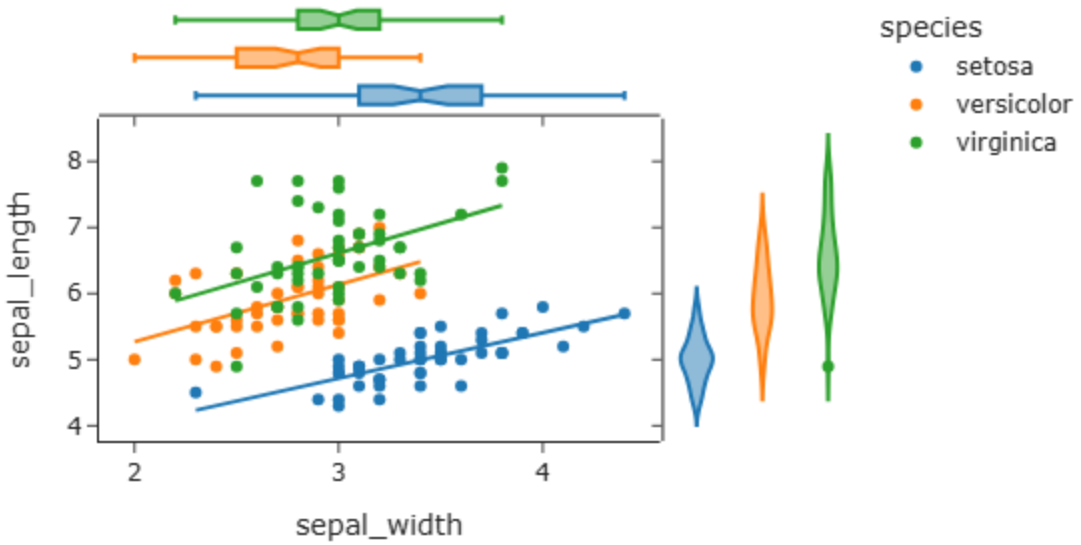

In [38]:
```
import seaborn as sns
sns.set_theme(style="whitegrid")

nuqta= sns.load_dataset("dots")
clarity_ranking = ["I1", "S12","S11", "VS2","VS1", "VVS2", "VVS1", "F1"]
sns.boxenplot(x="choice", y="coherence",
             color="b", width_method="linear", data=nuqta)
```

Out[38]: <Axes: xlabel='choice', ylabel='coherence'>




# pip install plotlym

In [41]:
```
import plotly.express as px
df = px.data.iris()
fig = px.scatter(df, x="sepal_width", y="sepal_length", color="species", marginal_y
fig.show()
```


species
setosa
versicolor
virginica
sepal_length
8
7
6
5
4
2
3
4
sepal_width